

Time Series Forecasting - Complete Study Notes

1. Data Loading

Data Loading is the process of importing data from a CSV file, database, or API into Python so it can be analyzed. Pandas is commonly used through `pd.read_csv()`.

2. Data Cleaning

Data Cleaning is the process of correcting or removing inaccurate, duplicate, missing, or invalid data. Clean data improves the quality of analysis and forecasting.

3. Feature Engineering

Feature Engineering is the process of creating new useful variables from existing columns.

Example: $\text{Total_Sales} = \text{Quantity} \times \text{UnitPrice}$.

4. Time Series Creation

Time Series Creation organizes data by time (daily, weekly, monthly, etc.) so trends and patterns can be analyzed and forecasted.

5. Seasonal Decomposition

Seasonal Decomposition separates data into Trend, Seasonality, and Residual components to understand hidden patterns.

6. Prophet

Prophet is a forecasting library developed by Meta. It models trend, seasonality, and holiday effects and is widely used for business forecasting.

7. SARIMAX

SARIMAX is a statistical forecasting model that captures trend, seasonality, and external variables for accurate time-series prediction.

8. LSTM

LSTM (Long Short-Term Memory) is a deep learning neural network designed to learn long-term dependencies in sequential data such as sales, stock prices, and weather.

9. Evaluation

Evaluation measures model performance using metrics such as MAE (Mean Absolute Error) and RMSE (Root Mean Squared Error). Lower values indicate better predictions.

10. Portfolio Learning Path

Recommended learning order: Pandas → Data Cleaning → EDA → Feature Engineering → Time Series Analysis → Seasonal Decomposition → Prophet → SARIMAX → LSTM → Real Projects.

Learning Flowchart

Pandas



Data Cleaning



EDA



Feature Engineering



Time Series Analysis



Seasonal Decomposition



Prophet



SARIMAX



LSTM



Forecasting Projects